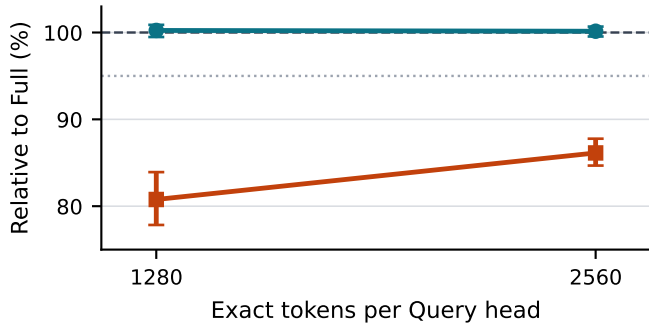
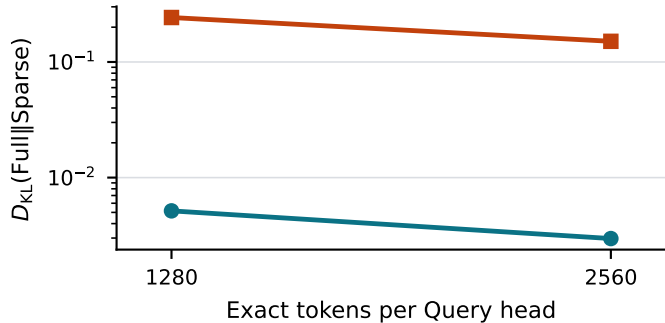


(a) Causal-PPL retention



(b) Distribution error



---- Full     Exact FP16 QK oracle     QK-balanced low-bit proxy